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Design and Simulation of a Pulse-Doppler Radar System

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3rd academic year

Abstract

This project demonstrates the design and simulation of a Pulse-Doppler Radar system using MATLAB. The system focuses on detecting a moving target by transmitting Linear Frequency Modulation (LFM) pulses, also known as "Chirps." The core objective is to accurately estimate the target's Range and Velocity using advanced signal processing techniques, including Matched Filtering for pulse compression and 2D-FFT for Doppler analysis. The final output is visualized through a Range-Doppler Map, ensuring high-precision detection even in the presence of noise.

Project Description

1. Signal Generation (The Chirp)

The radar starts by transmitting a Chirp signal, where the frequency increases linearly over time. This allows the system to send high-energy pulses to reach long distances while maintaining the ability to resolve targets clearly.

2. The Doppler Effect

When the transmitted signal hits a moving target, it returns with a frequency shift called the Doppler Shift.

- If the target is moving towards the radar, the frequency increases.
- If the target is moving away, the frequency decreases. This shift is what allows the system to calculate the exact Velocity of the target.

3. Signal Processing (Matched Filter)

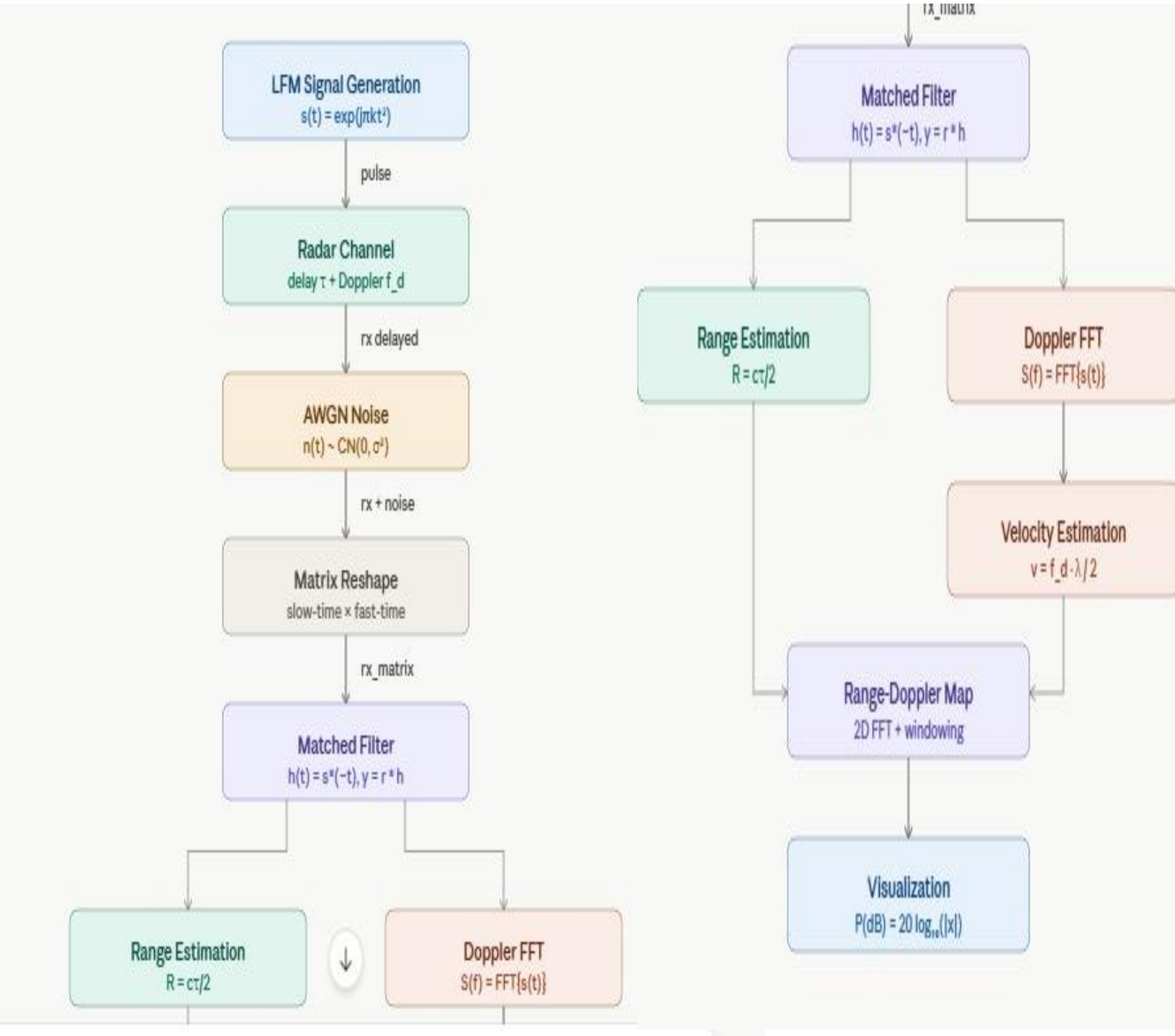
The received echo is passed through a Matched Filter, which performs two critical tasks:

- Pulse Compression: It gathers all the energy of the long pulse into one sharp, narrow peak (Pulse Compression), making the target appear as a precise point on the screen.
- SNR Improvement: It filters out the noise and maximizes the signal strength to ensure the target is clearly visible.

4. Final Output (Range-Doppler Map)

By applying a 2D-FFT, the system generates a map where the horizontal axis represents the Distance (Range) and the vertical axis represents the Speed (Velocity). The highest peak on this map indicates the target's real-time position and motion.

System Block Diagram



Mathematical Modeling and Radar Equations

(1) LFM Signal Generation: The transmitted signal is generated using: $s(t) = \exp(j\pi k t^2)$

where

$$s(t) = Rect\left(\frac{t}{\tau}\right)e^{j\pi\mu t^2} \tag{3.35}$$

is the complex envelope of $s_1(t)$.

(2) Radar Channel Model: The received signal includes: - Time delay: $\tau = 2R / c$

1.2. Range

Figure 1.1 shows a simplified pulsed radar block diagram. The time control box generates the synchronization timing signals required throughout the system. A modulated signal is generated and sent to the antenna by the modulator/transmitter block. Switching the antenna between the transmitting and receiving modes is controlled by the duplexer. The duplexer allows one antenna to be used to both transmit and receive. During transmission it directs the radar electromagnetic energy towards the antenna. Alternatively, on reception, it directs the received radar echoes to the receiver. The receiver amplifies the radar returns and prepares them for signal processing. Extraction of target information is performed by the signal processor block. The target's range, R, is computed by measuring the time delay, Δt, it takes a pulse to travel the two-way path between the radar and the target. Since electromagnetic waves travel at the speed of light, $c = 3 \times 10^8 m/\text{sec}$, then

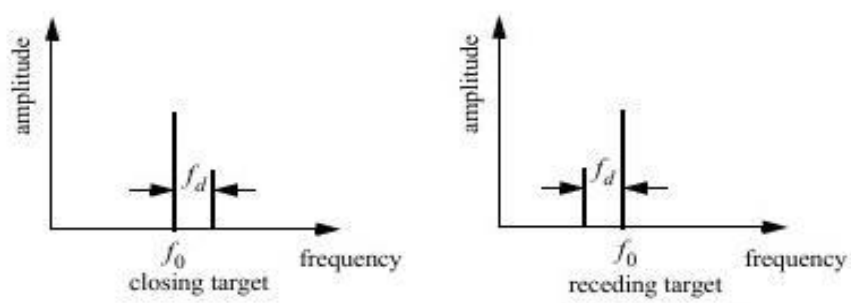
$$R = \frac{c\Delta t}{2} \tag{1.1}$$

where R is in meters and Δt is in seconds. The factor of $\frac{1}{2}$ is needed to account for the two-way time delay.

- Doppler shift: $f_d = 2v / \lambda$

$$f_d = \frac{2v}{c} f_0 = \frac{2v}{\lambda} \tag{1.35}$$

which is the same as Eq. (1.23). It can be shown that for a receding target the Doppler shift is $f_d = -2v/\lambda$. This is illustrated in Fig. 1.9.



- Received signal: r(t) = s(t-τ)·exp(j2πf_d t)

$$f_d = (\gamma - 1) f_0 \tag{3.109}$$

Using the approximation $\gamma \approx 1$ and Eq. (3.109), Eq. (3.107) is rewritten as

$$s_r(t) \approx e^{j2\pi f_d (t - t_0)} \, s(t - t_0) \tag{3.110}$$

where

$$s(t - t_0) = e^{-j2\pi f_0 t} s_1(t - t_0) \tag{3.111}$$

(3) Noise Model: Additive White Gaussian Noise (AWGN): n(t) ~ CN(0, σ^2)

where the noise quadrature components $n_I(t)$ and $n_Q(t)$ are uncorrelated zero mean low pass Gaussian noise with equal variances, ψ^2. The joint Probability Density Function (*pdf*) of the two random variables $n_I; n_Q$ is

$$f(n_I, n_Q) = \frac{1}{2\pi\psi^2} \exp\left(-\frac{n_I^2 + n_Q^2}{2\psi^2} \right) \tag{2.4}$$

$$= \frac{1}{2\pi\psi^2} \exp\left(-\frac{(r\cos\varphi - A)^2 + (r\sin\varphi)^2}{2\psi^2} \right)$$

(4) Matched Filtering: Optimal detection filter: h(t) = s*(-t)

$s_1(t)$ is given in Eq. (3.101). The matched filter response is given by the convolution integral

$$s_o(t) = \int_{-\infty}^{\infty} h(u) s_r(t - u) du \tag{3.112}$$

For a non-causal matched filter the impulse response $h(u)$ is equal to $s^*(-t)$; it follows that

$$s_o(t) = \int_{-\infty}^{\infty} s^*(-u) s_r(t - u) du \tag{3.113}$$

Substituting Eq. (3.111) into Eq. (3.113), and performing some algebraic manipulations, we get

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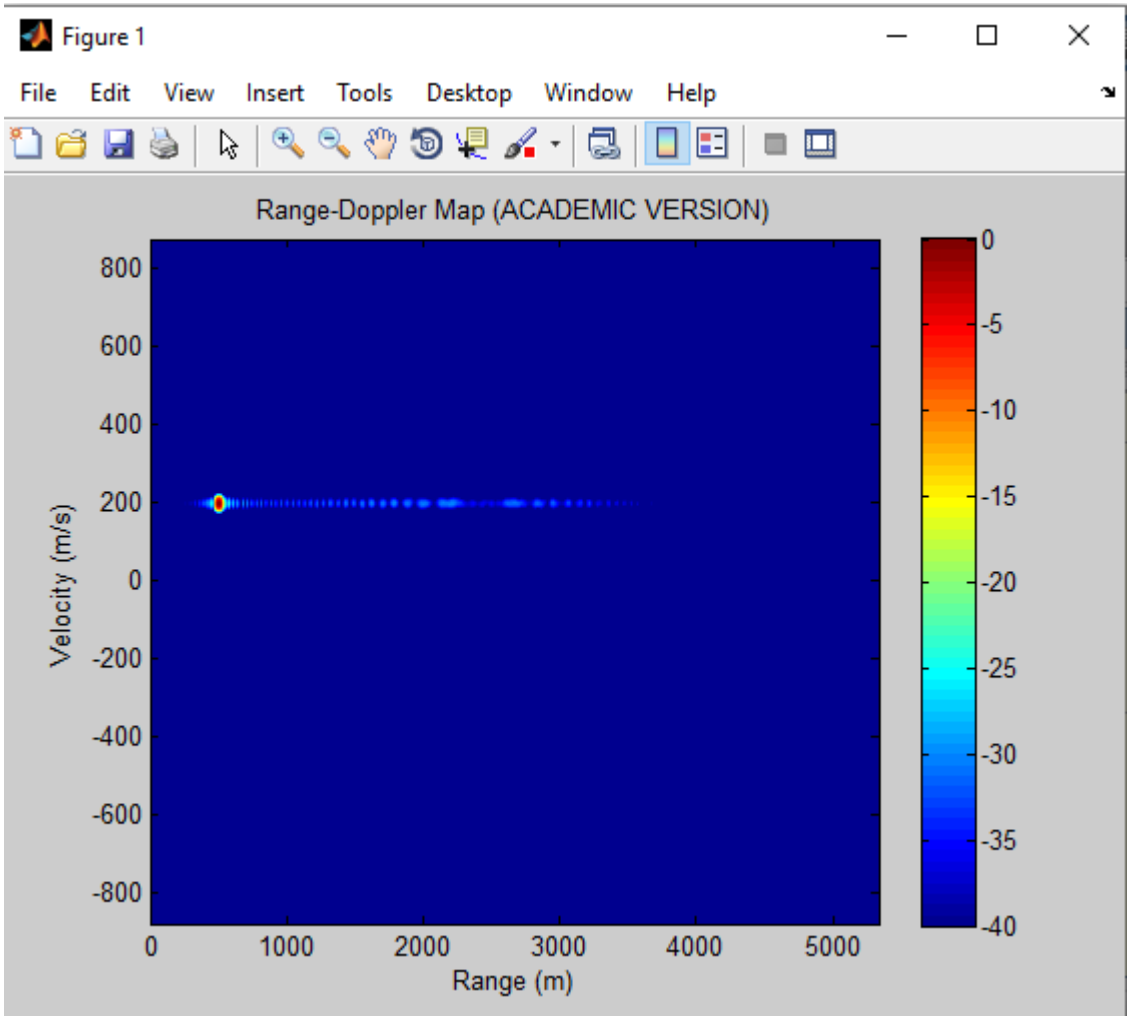
(5) Visualization: Power in dB: P(dB) = 20 log10(|x|)



Radar System Parameters

1. Range Resolution:

Explanation:

Range resolution refers to the ability of the radar system to distinguish between two targets that are at different distances from the radar. In this simulation, it is determined by the bandwidth of the transmitted signal.

Equation:
Range Resolution = c / 2B

where:
- c is the speed of light,
- B is the bandwidth of the transmitted signal (in this case, B = 5 x 10^6 Hz).

This formula is not explicitly written in the code but is derived from the system parameters.

2. Velocity Resolution:

Explanation:

Velocity resolution refers to the ability of the radar system to distinguish between two targets that are moving at different velocities. This is defined by the Doppler shift and depends on the pulse repetition frequency (PRF) and the FFT resolution.

Equation**:**

Velocity Resolution = λ / 2 * T_PRI

where**:**
- λ is the wavelength of the radar signal,
- T_PRI is the pulse repetition interval.

In the code, the velocity is estimated using Doppler frequency f_d, which depends on the velocity.

3. Signal-to-Noise Ratio (SNR):

Explanation:

SNR measures the ratio of the power of the desired signal to the power of the background noise. A higher SNR indicates a clearer signal.

Formula**:**

SNR is typically calculated as:

SNR = Signal Power / Noise Power

In the code, noise is added using an additive white Gaussian noise (AWGN) model with a specified noise power. SNR would depend on the power of the signal and the noise power, which is manually set to 0.01.

4. Peak Doppler (PD):

Explanation:

PD refers to the maximum Doppler shift or frequency corresponding to the maximum velocity of the target. This is estimated through the FFT of the Doppler-shifted signal.

Equation:

The velocity is estimated as:
$v = f_d * \lambda / 2$
where f_d is the Doppler frequency and λ is the wavelength. This is used in the code to estimate velocity based on the maximum Doppler frequency.

5. Probability of False Alarm (PFA)

Definition

Probability of False Alarm (PFA) is the probability that a system detects a target or signal when no actual target exists.

In other words, the system makes a false decision because of noise or interference.

Example — Radar System

- The radar declares that a target is present.
- In reality, there is no target.
- → False AlarmThis erroneous declaration is called a

Mathematical Expression

Formula: $P_{FA} = P(V_T > V_{th} \mid H_o)$

Where:

- V_T → Received signal voltage
- V_{th} → Detection threshold
- H_o → Hypothesis of no target present

This represents the probability that noise alone exceeds the detection threshold.

Key Relationships

> ⬆ Threshold increases → PFA decreases
> ⬇ Threshold decreases → PFA increases

6.Ambiguity

Definition

Ambiguity is the inability of a system to determine the exact value of a parameter because more than one possible interpretation exists. It commonly appears in radar and communication systems.

Example — Radar System

If pulses are transmitted too close together, the radar may not determine which reflected pulse belongs to which transmitted pulse — creating ambiguity.

A) Range Ambiguity

Occurs when the radar cannot determine the exact distance of the target.

```
Formula:  R_max  =  (c × T_r) / 2
```

Where:

- R_max → Maximum unambiguous range
- c → Speed of light
- T_r → Pulse repetition time

B) Doppler Ambiguity

Occurs when the radar cannot determine the exact velocity of the target.

```
Formula:  f_d  =  (2 × v) / λ
```

Where:

- f_d → Doppler frequency
- v → Target velocity
- λ → Wavelength

> 📌 Ambiguity = Uncertainty in measurement due to multiple possible valid values.

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7. Latency

Definition

Latency is the time delay between sending a request or signal and receiving the response. It represents the waiting time in a system.

Example

If data is sent now and received after 10 ms → Latency = 10 ms

General Formula

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Formula: Latency = T_response − T_request
```

Where:

- T_response → Time at which the response is received
- T_request → Time at which the request was sent

Network Latency — Component Breakdown

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Formula: Total Latency = Processing + Transmission + Propagation + Queueing
```

Component	Description
Processing Delay	Time required to process the data
Transmission Delay	Time required to send data over the medium
Propagation Delay	Time required for the signal to travel from source to destination
Queueing Delay	Waiting time in queue before transmission begins

> ✅ Lower latency → Faster system performance
> ❌ Higher latency → Slower system performance